Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:
This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. The modular design of the VBCUA application (separate transcription, scoring, audio-analysis, and UI components) supports maintainability and future extension.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	
2	Proper File Structure	Files and folders are logically organized	Yes	Separate modules for transcription, scoring, audio analysis, and UI
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	
4	Function / Method Names	Functions are descriptively named	Yes	
5	Code Comments	Inline and block comments explain logic	Yes	
6	Modular Design	Code is split into reusable functions/modules	Yes	Whisper, Sentence-BERT, and Librosa logic kept in separate functions
7	No Redundant Code	Duplicate or unused code is removed	Yes	
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Handles missing/invalid audio files and empty transcriptions

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	AudioProcessor	Python (Librosa)	Fluency analysis & waveform generation	High
2	TranscriptionEngine	Python (OpenAI Whisper)	Speech-to-text conversion	High
3	SimilarityScorer	Python (Sentence-Transformers)	Semantic comparison against reference concepts	High
4	ReportGenerator	Python (ReportLab)	PDF report creation and export	Medium
5	UIComponents	Python (Streamlit)	Shared across dashboard pages and result views	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Modules are clearly separated by responsibility (audio, NLP, UI, reporting)
Readability	4	Clear naming and comments; a few dense audio-processing sections could use more inline notes
Reusability	4	Core processing modules can be reused in other speech-analysis projects
Documentation / Comments	4	Well documented overall, setup instructions included
Overall Score	4.25	